Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Multivariate Time Series I

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VARMA Model

Vector Autoregressive Moving Average (VARMA) model

A VARMA(p, q) model

$$\mathbf{X}_t = \boldsymbol{\mu} + \sum_{i=1}^{p} \boldsymbol{\Phi}_i \mathbf{X}_{t-i} + \sum_{j=1}^{q} \boldsymbol{\Theta}_j \boldsymbol{\varepsilon}_{t-j} + \boldsymbol{\varepsilon}_t$$

where $\boldsymbol{\mu}$ is mean vector, $\boldsymbol{\Phi}_i$ is AR matrices, $\boldsymbol{\Theta}_j$ is MA matrices, and $\boldsymbol{\varepsilon}_t$ is a white noise.

K-dimensional vector time series $\mathbf{X}_t = (\mathbf{X}_t, \ldots, \mathbf{X}_{tK})'$ where $\mathbf{X}_{tk}$ represent the return at time t of $\mathbf{X}_k$ ($1 \leq k \leq K$).

Stationarity and Vector White Noise

- If the vector time series $\mathbf{X}_t = (\mathbf{X}_t, \ldots, \mathbf{X}_{tK})'$ is stationary, then each of its component series $\mathbf{X}_{tk}$ (k = 1, …, K) is a univariate stationary time series.
- The vector time series $\varepsilon_t = (\varepsilon_t, \ldots, \varepsilon_{tK})'$ is called a vector white noise, the vector white noise is clearly stationary, and each of its components is a univariate white noise series.

Granger Causality and Impulse Response Analysis

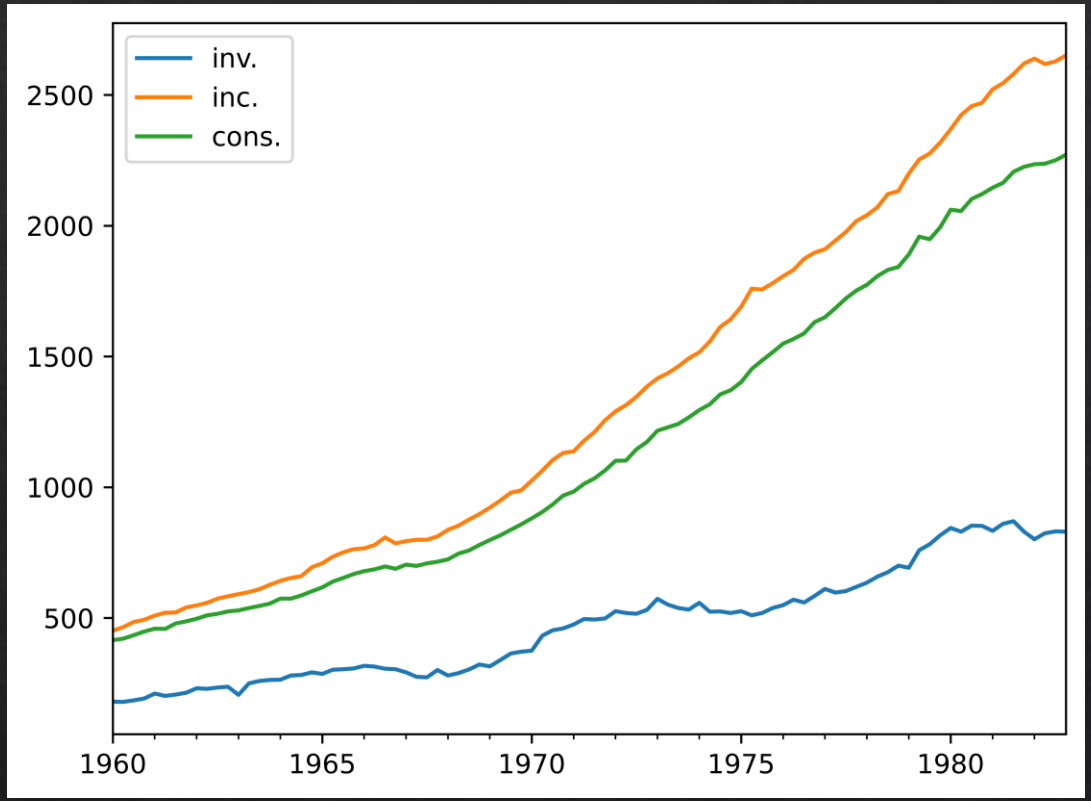
- The model implies the existence of *Granger causality* between two series with $\mathbf{X}_{t1}$causing $\mathbf{X}_{t2}$, but not being caused by $\mathbf{X}_{t2}$.
- From another perspective, we will investigate the relationship between variables: in other words, we will examine the *response* of one variable to an *impulse* in another variable in the system.

Dataset Description

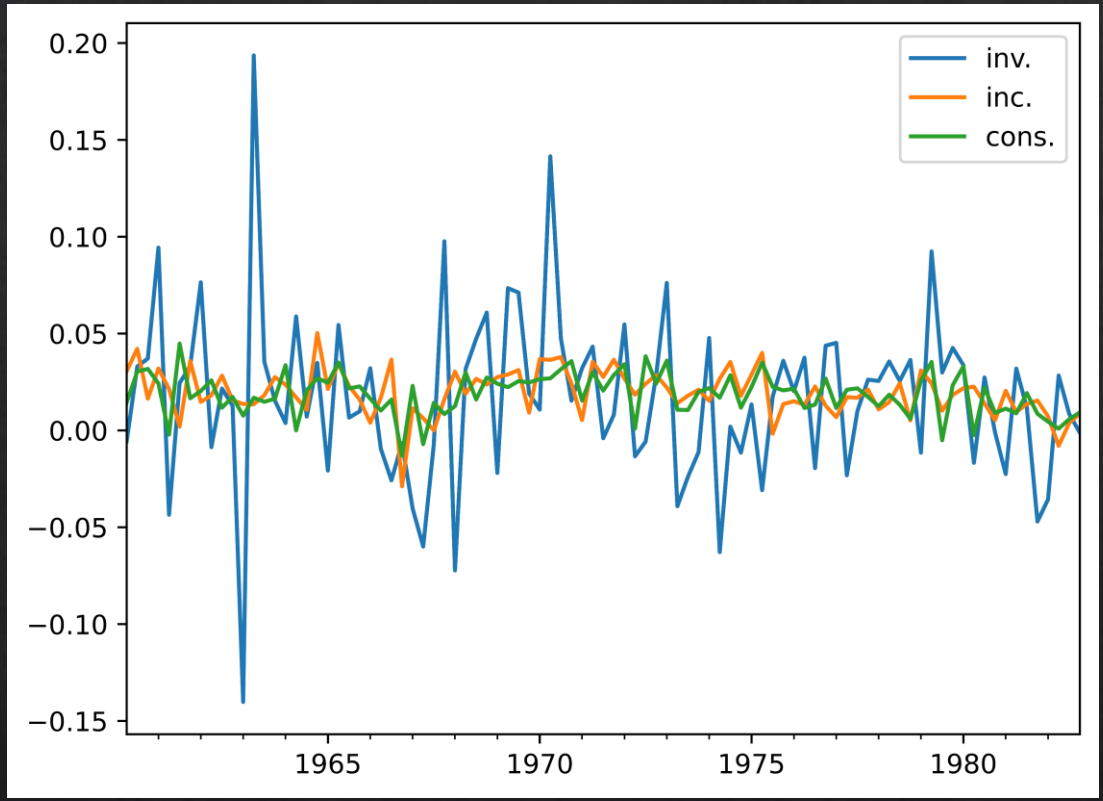
West German Macroeconomic Data

- The data is quarterly, seasonally adjusted.
- There are three variables, investment (inv.), disposable income (inc.) and consumption expenditures (cons.) from the first quarter of 1960 to the fourth quarter of 1982.

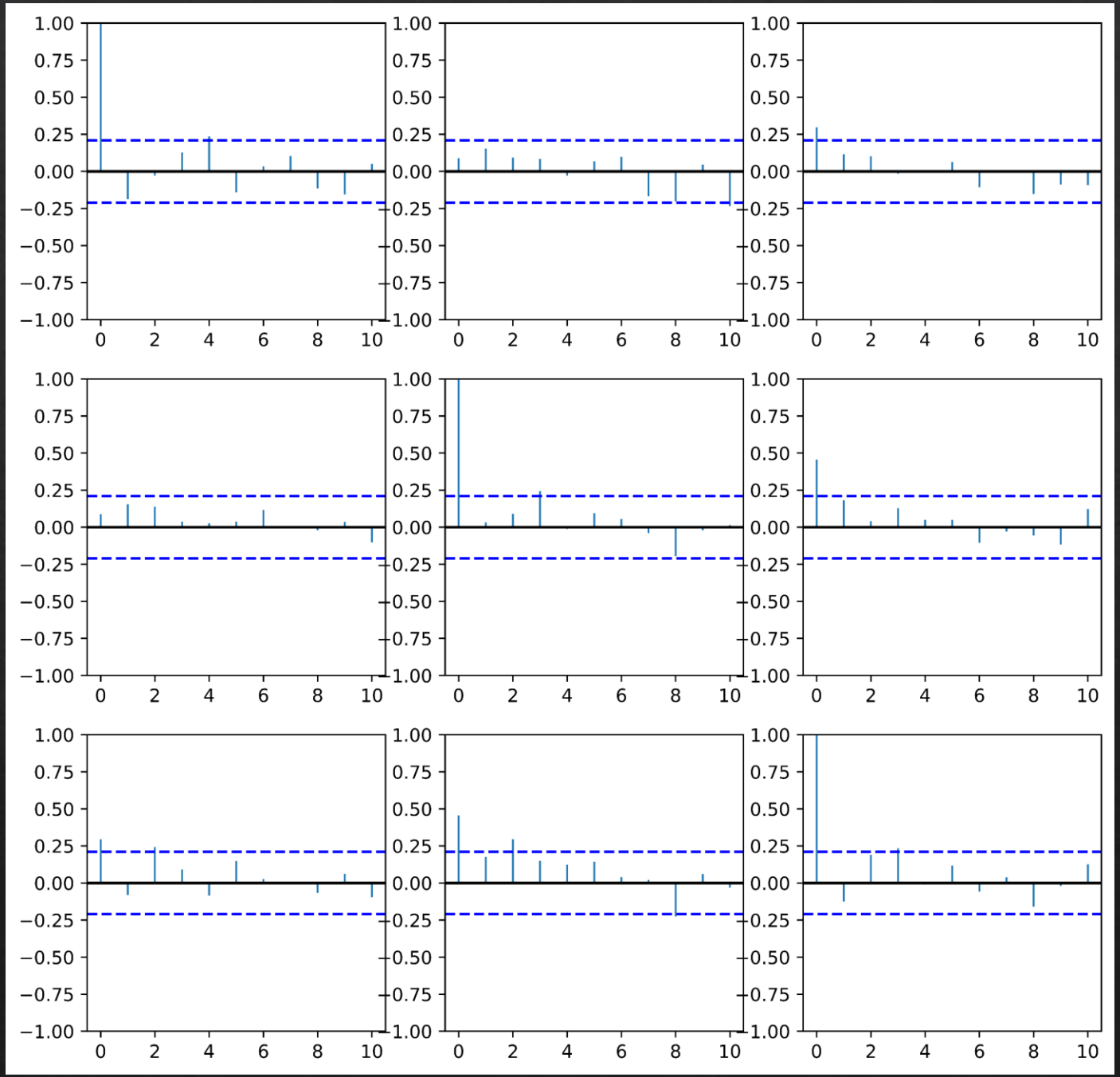
Time Series Plot



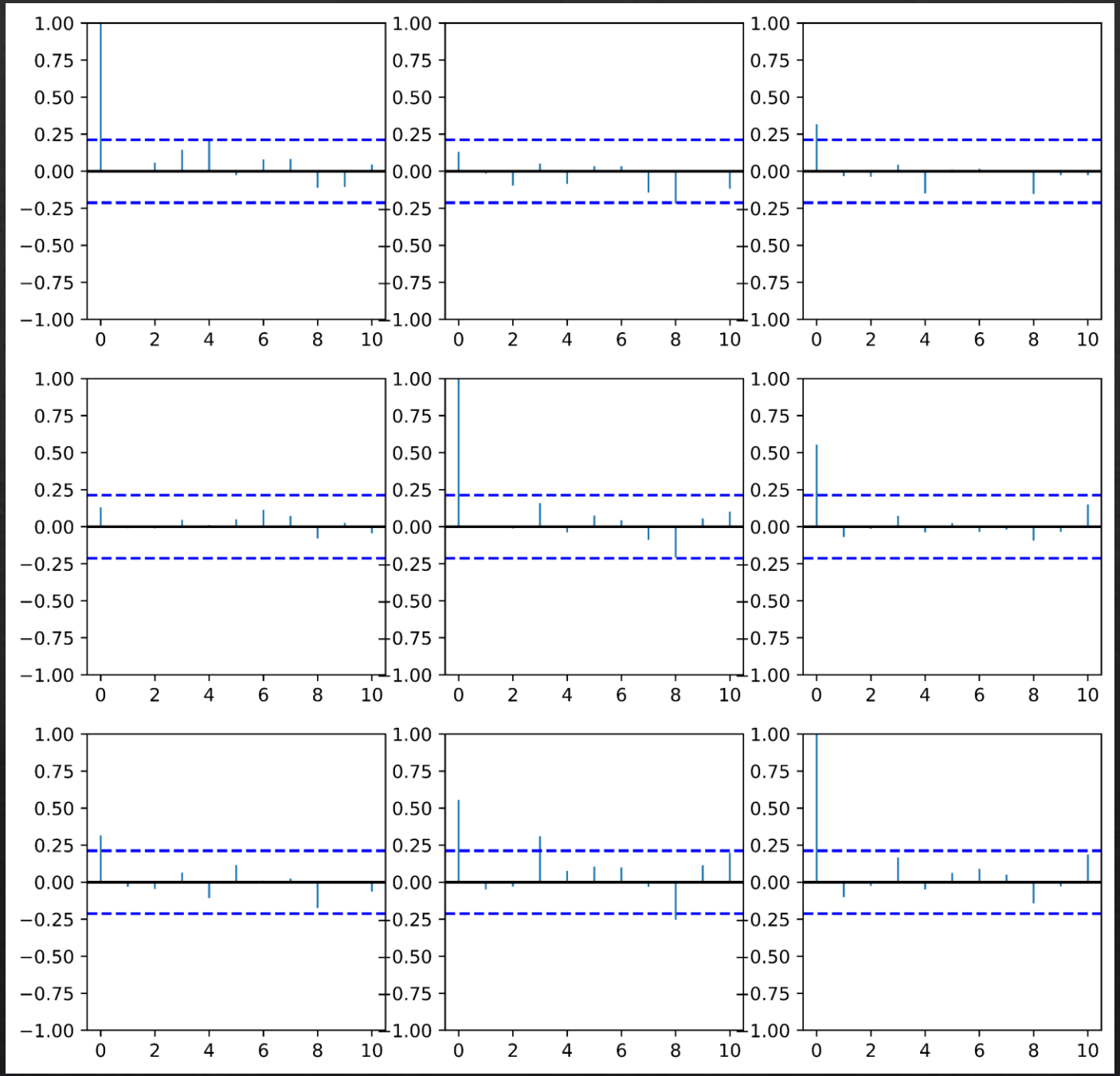
First Difference of Logarithms



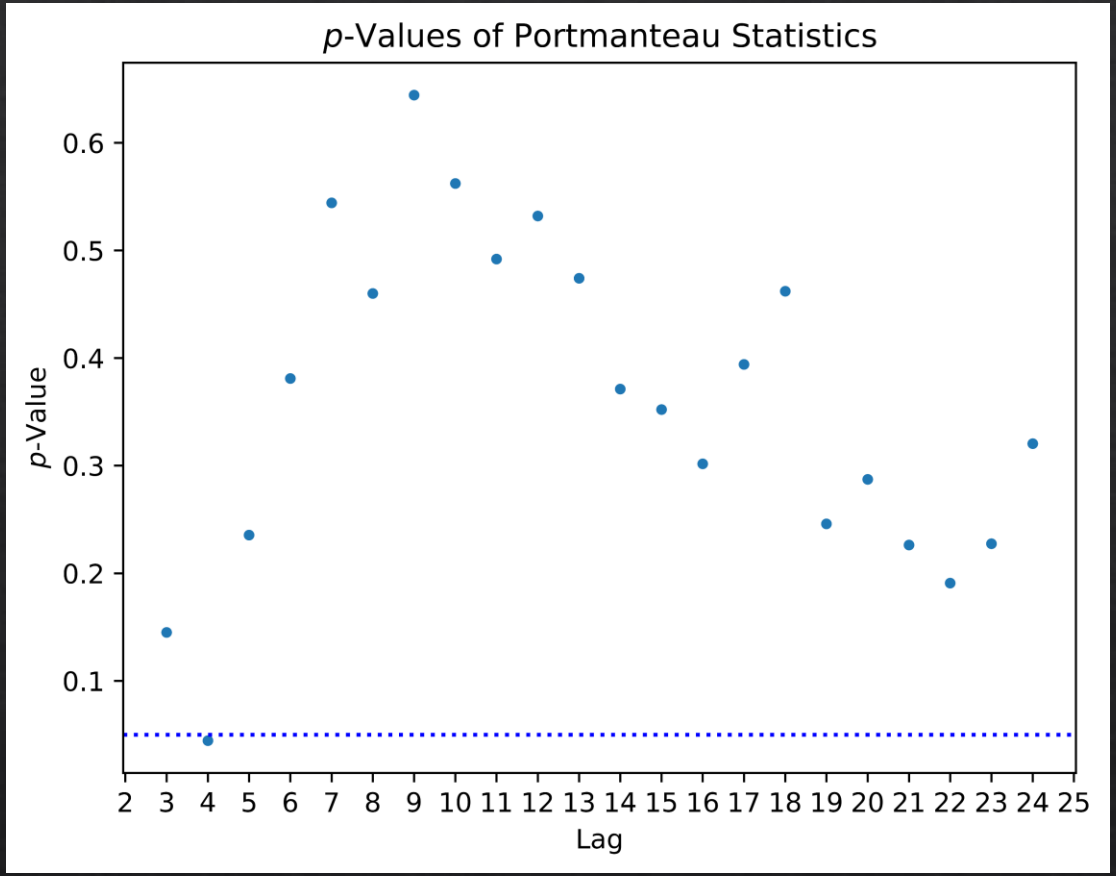
Correlation Plot of first differences of logs



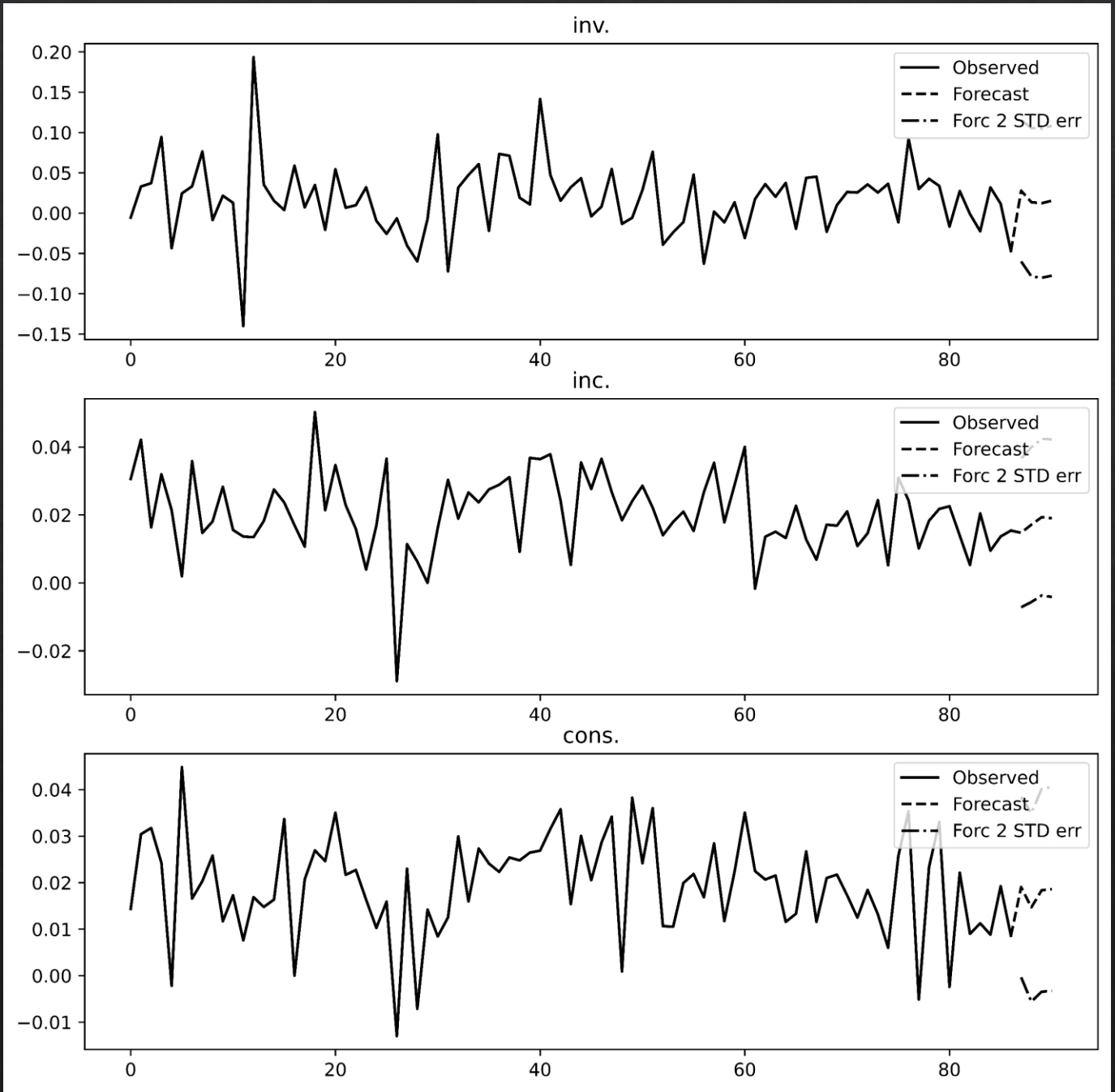
Correlation Plot of model residual



p-values plot of the model residual



Forecast Future values with VAR(2) model



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VAR(2) model

The model equation can write as follows:

$$\begin{bmatrix} X_{t1} \\ X_{t2} \\ X_{t3} \end{bmatrix} = \begin{bmatrix} -0.011 \\ 0.015 \\ 0.014 \end{bmatrix} + \begin{bmatrix} -0.029 & 0.379 & 0.665 \\ 0.039 & -0.146 & 0.256 \\ -0.005 & 0.281 & -0.321 \end{bmatrix} \begin{bmatrix} X_{t-1,1} \\ X_{t-1,2} \\ X_{t-1,3} \end{bmatrix} + \begin{bmatrix} -0.138 & 0.182 & 0.668 \\ 0.053 & 0.007 & 0.018 \\ 0.041 & 0.363 & -0.128 \end{bmatrix} \begin{bmatrix} X_{t-2,1} \\ X_{t-2,2} \\ X_{t-2,3} \end{bmatrix} + \begin{bmatrix} \varepsilon_{t1} \\ \varepsilon_{t2} \\ \varepsilon_{t3} \end{bmatrix}$$

The covariance matrix of residuals as follows:

$$\begin{bmatrix} 0.002004 & 0.000065 & 0.000140 \\ 0.000065 & 0.000125 & 0.000061 \\ 0.000140 & 0.000061 & 0.000098 \end{bmatrix}$$

Granger Causality

Hypothesis

- H_0 :income does NOT affect consumption
- H_1 :income affects consumption

p-value: 0.001

Income Grange-causes consumption.

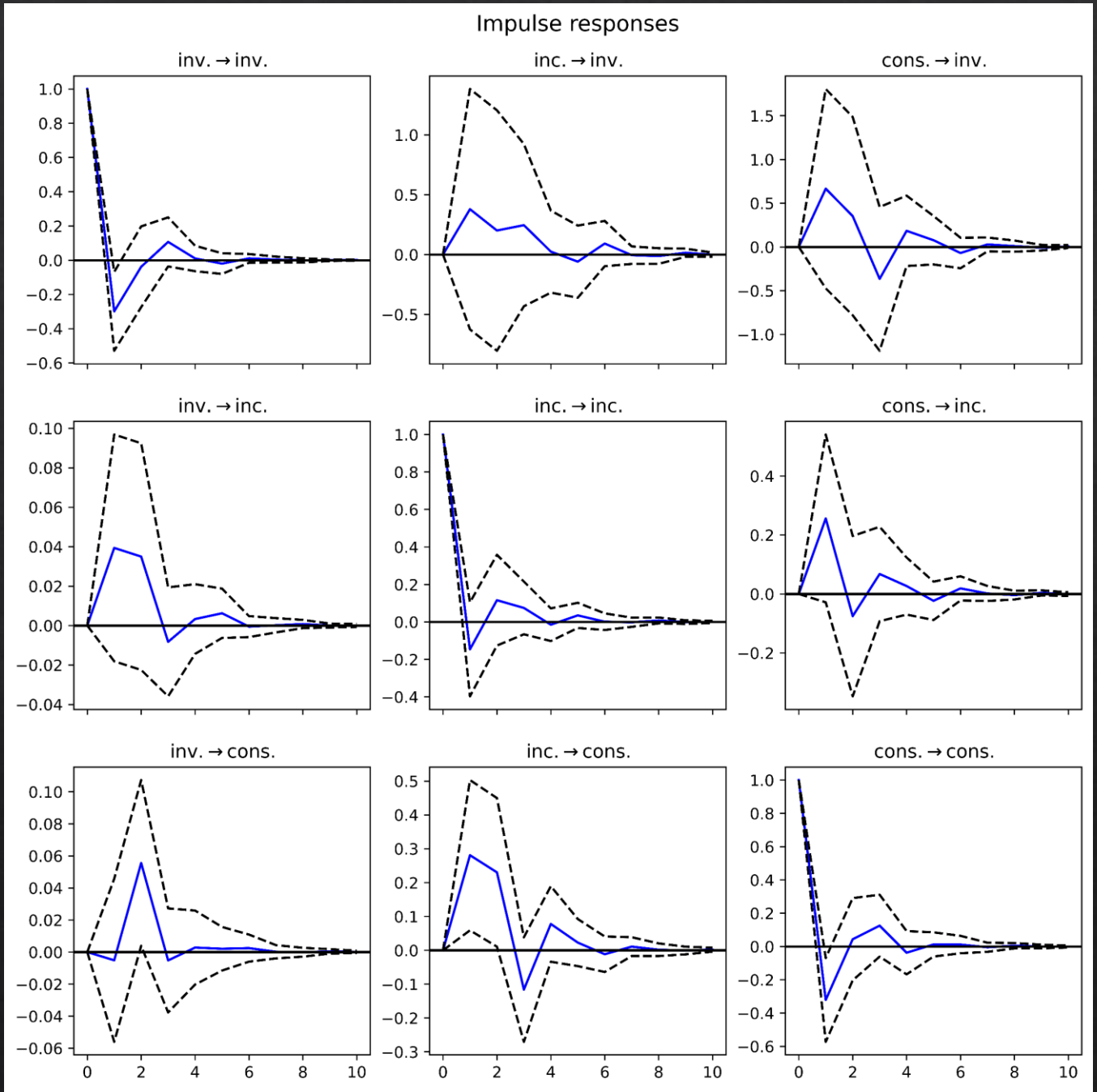
Hypothesis

- H_0 :Consumption does NOT affect income
- H_1 :Consumption affects income

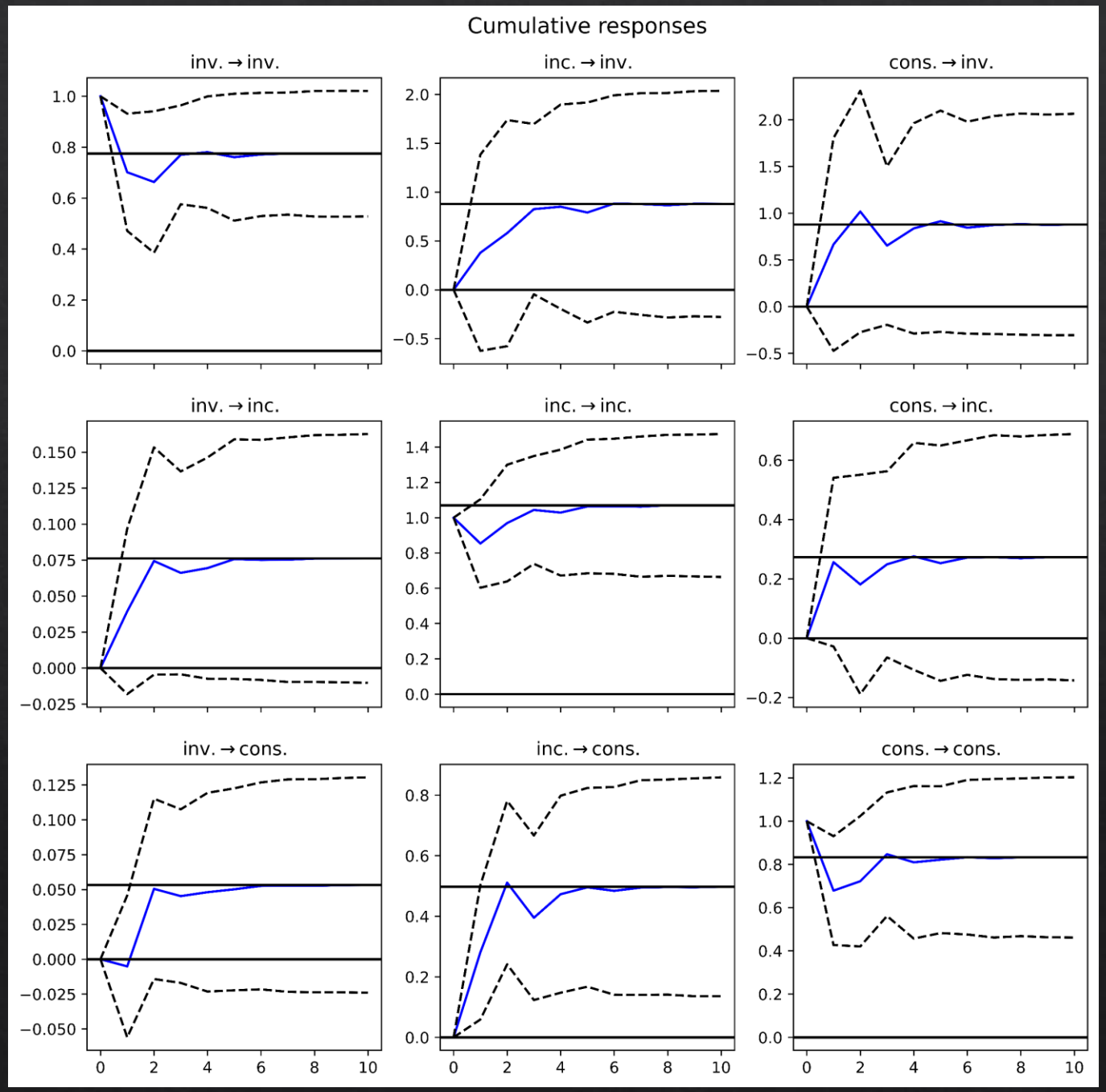
p-value: 0.182

Consumption does not Grange-causes income.

Impulse Response



Cumulative Response



Summary

- VAR(2) model performs well, capturing the relationships and providing reasonable forecasts for the next 4 points.
- Granger causality shows a one-way effect from income to consumption, with no significant reverse relationship.
- Impulse response results indicate that income shocks have the strongest and most persistent impact on the system.